Coordinates in 2D

Algebra · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

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Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 97 minutes · Total marks: 97

KEY RULES

Midpoint	$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
Length	$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
Gradient	$m=\frac{y_2-y_1}{x_2-x_1}$
Sketch	Mark the points and draw the shape to check

WORKED EXAMPLE 1

[2 marks]

Find the midpoint of $A(2, 5)$ and $B(8, -1)$.

Step 1	Add the x values and halve	$\frac{2+8}{2}=5$
Step 2	Add the y values and halve	$\frac{5+(-1)}{2}=2$

Answer: $(5, 2)$

WORKED EXAMPLE 2

[3 marks]

Work out the length of AB where $A=(1, 1)$ and $B=(5, 4)$.

Step 1	Differences	4 and 3
Step 2	Pythagoras	$\sqrt{4^2+3^2}=\sqrt{25}$

Answer: 5

WORKED EXAMPLE 3

[3 marks]

$M(4, 7)$ is the midpoint of $A(1, 3)$ and B. Find the coordinates of B.

Step 1	x: the midpoint is 3 units right of A, so go 3 more	$4+3=7$
Step 2	y: the midpoint is 4 units above A, so go 4 more	$7+4=11$

Answer: $B=(7, 11)$

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1 Write down the equation of the horizontal line through $(3, 7)$.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Write down the equation of the vertical line through $(-2, 5)$.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

3 Work out the length of AB where $A = (-3, 2)$ and $B = (5, -4)$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Find where the line $y = 3x - 6$ crosses the x-axis and the y-axis.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Find where the line $2x + 5y = 20$ crosses the axes.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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Coordinates in 2D

6 1 The point P has coordinates (3, 4)
The point Q has coordinates (9, 16)

M is the midpoint of the line PQ

Find the coordinates of M

(.....,.....)

(Total for Question 1 is 2 marks)

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7 Point A has coordinates $(-4, 9)$
Point B has coordinates $(1, 5)$

Find the coordinates of the midpoint of AB.

(.....,.....)

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8 Point A has coordinates $(-3, 11)$
Point B has coordinates $(47, b)$
The midpoint of AB has coordinates $(a, -19)$

Find the value of a and the value of b.

$a =$ .....

$b =$ .....

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 6

9 B is the point with coordinates $(1, 4)$
C is the point with coordinates $(6, 9)$

Find the coordinates of the midpoint of BC.

(..... , ...)

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10 6 The points A and B are on a coordinate grid.

The coordinates of A are $(6, 4)$
The coordinates of B are $(17, j)$ where j is a constant.

The midpoint of AB has coordinates $(k, 15)$ where k is a constant.

Find the value of j and the value of k

$j =$ .....

$k =$ .....

(Total for Question 6 is 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2023 · Question 6

11 A has coordinates $(11, 3e)$
B has coordinates $(1, 7e)$
The midpoint of AB has coordinates (x, y)

(a) Find the value of x.

$x =$ .....

(1)

(b) Find an expression for y in terms of e.
Simplify your answer.

$y =$ .....

(2)

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12 Two circles, C_1 and C_2, are drawn on a centimetre grid, with a scale of 1 cm for 1 unit on each axis.

The centre of circle C_1 is at the point with coordinates (–1, 3) and the radius of C_1 is 13 cm.

The centre of circle C_2 is at the point with coordinates (7, 18) and the radius of C_2 is 6 cm.

(a) Work out the distance between the centre of C_1 and the centre of C_2

..... cm

(3)

Explain why circle C_1 intersects circle C_2

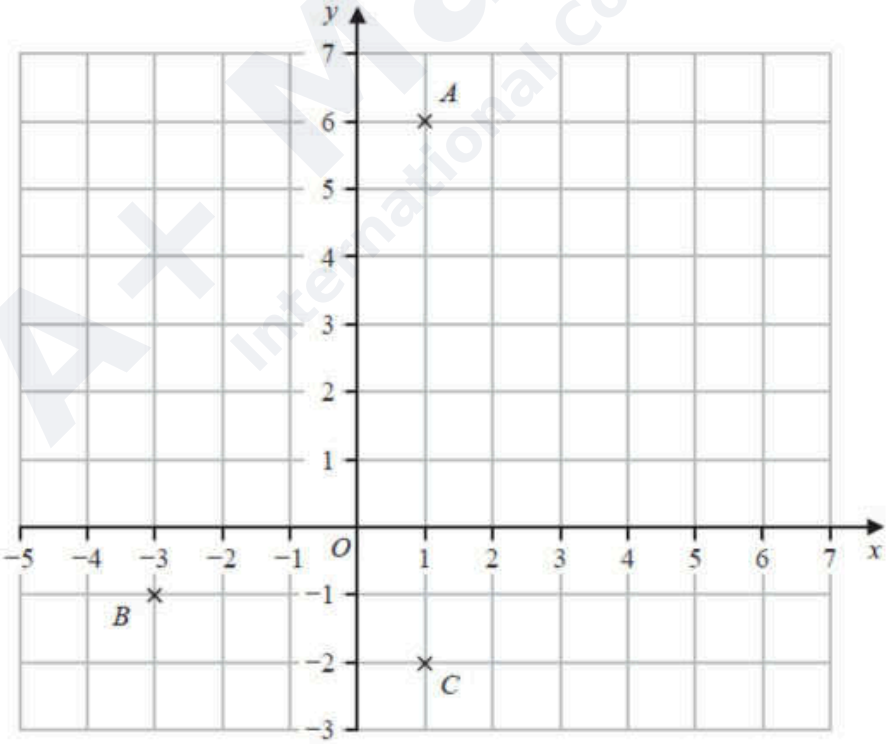
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(1)

(Total for Question 10 is 4 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 10

13 The diagram shows three points, *A*, *B* and *C*, marked on a grid.



(a) Write down the coordinates of point *A*

(..... , ...)

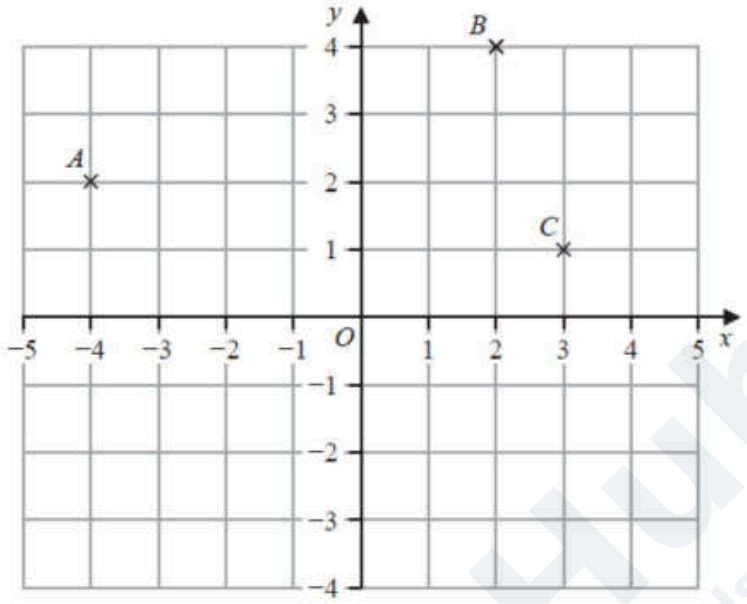
(1)

(b) Write down the coordinates of point *B*

(..... , ...)

Coordinates in 2D

14 The diagram shows three points, A, B and C, marked on a grid.



(a) Write down the coordinates of A

(..... , ...)

(1)

(b) Find the coordinates of the midpoint of AB

(..... , ...)

(2)

D is the point on the grid so that $ABCD$ is a rectangle.

(c) Find the coordinates of D

(..... , ...)

(2)

(Total for question = 5 marks)

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15 **22** ABC is a triangle in which angle $ABC = 90°$

p and q are integers such that

the coordinates of A are $(p, 10)$
the coordinates of B are $(-1, -5)$
the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

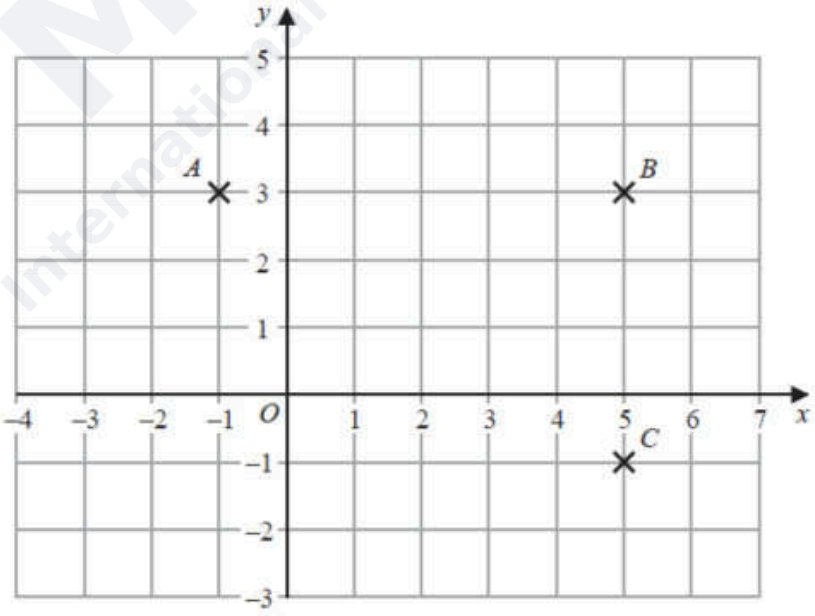
$p =$ .....

$q =$ .....

(Total for Question 22 is 5 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · January 2022 · Question 22

16 The three points A, B and C are marked on a centimetre grid.



(a) Write down the coordinates of A

(..... , ...)

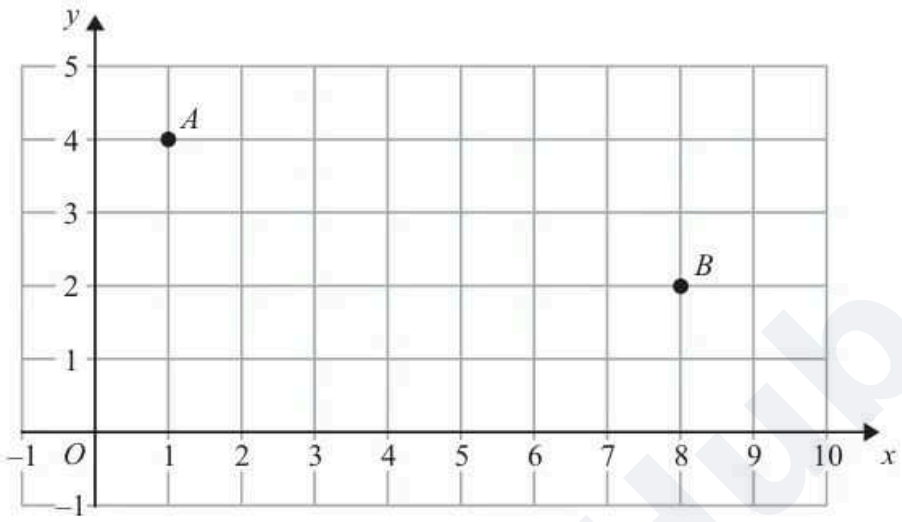
(1)

(b) Find the coordinates of the midpoint of BC

(..... , ...)

(2)

17 Two points, A and B, are plotted on a centimetre grid.
A has coordinates (1, 4) and B has coordinates (8, 2).



(a) Work out the coordinates of the midpoint of AB.

(.....,.....)

(2)

(b) Use Pythagoras' Theorem to work out the length of AB.
Give your answer correct to 3 significant figures.

..... cm

(4)

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18 **25** The straight line with equation $y - 2x = 7$ is the perpendicular bisector of the line AB where A is the point with coordinates $(j, 7)$ and B is the point with coordinates $(6, k)$

Find the coordinates of the midpoint of the line AB
Show clear algebraic working.

(..... , ...)

(Total for Question 25 is 6 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2023 · Question 25

19 **25** $ABCD$ is a trapezium with AB parallel to DC

A is the point with coordinates $(-4, 6)$
B is the point with coordinates $(2, 3)$
D is the point with coordinates $(-1, 8)$

The trapezium has one line of symmetry.
The line of symmetry intersects CD at the point E

Work out the coordinates of the point E

(..... , ...)

(Total for Question 25 is 6 marks)

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CHALLENGE QUESTIONS

20

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that

A has coordinates $(-4, -2)$
B has coordinates $(1, 10)$
C has coordinates $(19, a)$
D has coordinates (b, c)

(a) Work out the value of a, the value of b and the value of c.

$a =$ .....

$b =$ .....

$c =$ .....

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

..... cm

(3)

(Total for Question 18 is 7 marks)

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21 Triangle HJK is isosceles with $HJ = HK$ and $JK = \sqrt{80}$

H is the point with coordinates $(-4, 1)$
J is the point with coordinates $(j, 15)$ where $j < 0$
K is the point with coordinates $(6, k)$

M is the midpoint of JK.
The gradient of HM is 2

Find the value of j and the value of k.

$j =$ .....

$k =$ .....

(Total for question = 6 marks)

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Coordinates in 2D

22 $ABCD$ is a kite.

$AB = AD$ and $CB = CD$

The point B has coordinates $(k, 1)$ where k is a negative constant.
The point D has coordinates $(8, 7)$

The straight line **L** passes through the points B and D

The straight line **L** is parallel to the line with equation $5y - 3x = 6$

Find an equation of AC
Give your answer in the form $px + qy = r$ where p, q and r are integers.
Show your working clearly.

.....

(Total for question = 6 marks)

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23 **24** $ABCD$ is a kite with $AB = AD$ and $CB = CD$

A is the point with coordinates $(-2, 10)$

B is the point with coordinates $\left(-\frac{27}{5}, 4\right)$

C is the point with coordinates $(4, -5)$

Work out the coordinates of D

(.....,.....)

(Total for Question 24 is 6 marks)

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24 **22** $ABCD$ is a kite, with diagonals AC and BD, drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D.

(.....,.....)

(.....,.....)

(Total for Question 22 is 7 marks)

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TOTAL FOR PAPER IS 97 MARKS

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